

FIGURING DISTANCE, RATE, AND TIME

Distance, rate, and time are all related. If you travel faster, you can cover more distance in less time. You can travel 200 miles in a lot less time if you drive at 50 miles per hour than if you drive at 20 miles per hour.

$$\text{Rate} = \frac{\text{Distance}}{\text{Time}}$$

$$r = d/t$$

$$\text{Time} = \frac{\text{Distance}}{\text{Rate}}$$

$$t = d/r$$

$$\text{Distance} = \text{Rate} * \text{Time}$$

$$d = r * t$$

ONE
IMP
COMMENT

DO NOT
WRITE

R D T

Revenue

scratch

$$(t) r = d/t(t)$$

$$rt = d$$

$$\boxed{d = rt}$$

$$(t) r = d/t(t)$$

$$\frac{rt}{r} = \frac{d}{r}$$

$$t = d/r$$

INTEGRAL

DERIVATIVE

Distance

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$$\begin{aligned} \textcircled{1} \quad d &= r t \\ &= \left(\frac{50 \text{ mi}}{1 \text{ hr}} \right) (3.5 \text{ hrs}) \\ &= 50 (3.5) \text{ mi} \\ \boxed{d} &= \boxed{175 \text{ mi}} \end{aligned} \quad \begin{aligned} \textcircled{2} \quad r &= d/t \\ (3.5 \text{ hrs}) \frac{50 \text{ mi}}{1 \text{ hr}} &= \frac{d}{3.5 \text{ hrs}} (3.5 \text{ hrs}) \\ 3.5 (50) &= d \\ \boxed{d} &= \boxed{175 \text{ mi}} \end{aligned}$$

Distance = Rate * Time

The **distance formula** shows the relationship between distance, rate, and time.
distance (d) = rate (r) * time (t), or $d = rt$

If you know any two of the values in the distance formula, you can find the third by using one of the formulas shown in the following examples.

USING THE DISTANCE FORMULA

Example Rudy drove an average of 50 miles per hour for $3\frac{1}{2}$ hours. How far did he drive?

Step 1

Write the distance formula to find the distance.

$$d = rt$$

Step 2

Substitute the known values for rate and time.

$$d = 50 \times 3\frac{1}{2}$$

Step 3

Multiply to find the distance.

$$\frac{25}{1} \times \frac{7}{2} = 175 \text{ miles}$$

Rudy drove **175 miles** in $3\frac{1}{2}$ hours at 50 miles per hour.

Find the distance traveled in the problems below. Round to the nearest mile.

1A. rate = 45 mph

time = 4 hours

distance = _____ miles

$$r = d/t$$

$$\frac{45 \text{ mi}}{1 \text{ hr}} = \frac{d}{4 \text{ hrs}}$$

$$4 \text{ hrs} \left(\frac{45 \text{ mi}}{1 \text{ hr}} \right)$$

$$= \frac{d}{4 \text{ hrs}} (4 \text{ hrs})$$

$$4(45) = d$$

$$\boxed{d = 180 \text{ mi}}$$

1B. rate = 36 mph

time = $2\frac{1}{2}$ hours

distance = _____ miles

$$r = d/t$$

$$\frac{36 \text{ mi}}{1 \text{ hr}} = \frac{d}{2\frac{1}{2} \text{ hrs}}$$

$$\frac{36 \text{ mi}}{1 \text{ hr}} = \frac{d}{\frac{5}{2} \text{ hrs}}$$

$$\left(\frac{5}{2} \text{ hrs} \right) \frac{36 \text{ mi}}{1 \text{ hr}} = \frac{2}{5} \text{ hrs} (d)$$

$$90 \text{ mi} = d$$

$$\boxed{d = 90 \text{ mi}}$$

1C. rate = 55 mph

time = $1\frac{3}{4}$ hours

distance = _____ miles

$$r = d/t$$

$$\begin{aligned} \frac{55 \text{ mi}}{1 \text{ hr}} &= \frac{d}{1\frac{3}{4} \text{ hrs}} \\ &= \frac{d}{\frac{7}{4} \text{ hrs}} \end{aligned}$$

$$\left(\frac{7}{4} \text{ hrs} \right) \frac{55 \text{ mi}}{1 \text{ hr}} = d \left(\frac{7}{4} \text{ hrs} \right) \left(\frac{7}{4} \text{ hrs} \right)$$

scratch

$$\begin{array}{r} 3 \overline{) 55} \\ \underline{7} \\ 385 \end{array} \quad \begin{array}{r} 96 \\ 4 \overline{) 385} \\ \underline{36} \\ 25 \\ \underline{24} \\ 1 \end{array}$$

$$d = 216.25 \text{ miles}$$

1D. rate = 48 mph

time = $5\frac{1}{3}$ hours

distance = _____ miles

$$r = d/t$$

$$\frac{48 \text{ mi}}{1 \text{ hr}} = \frac{d}{5\frac{1}{3} \text{ hrs}}$$

$$\frac{48 \text{ mi}}{1 \text{ hr}} = \frac{d}{\frac{16}{3} \text{ hrs}}$$

$$\left(\frac{16}{3} \text{ hrs} \right) \left(\frac{48 \text{ mi}}{1 \text{ hr}} \right) = \frac{d}{\left(\frac{16}{3} \text{ hrs} \right)}$$

$$256 \text{ mi} = d$$

$$\boxed{d = 256 \text{ mi}}$$

$$\begin{aligned} d &= r t \\ &= 48 (5\frac{1}{3}) \\ &= 48 (16\frac{2}{3}) \\ &= 162 \end{aligned}$$

$$\boxed{d = 256 \text{ mi}}$$

$$\begin{array}{r} 20 \\ 24 \\ \hline 7 \end{array}$$

$$d = 96.25 \text{ miles}$$

$$\boxed{d = 96 \text{ miles}}$$

Rate

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$$t = 2\frac{1}{2} \text{ hrs} = \frac{5}{2} \text{ hrs}$$

$$d = 100 \text{ mi}$$

$$r = \frac{100 \text{ mi}}{\frac{5}{2} \text{ hrs}}$$

$$\text{Rate} = \frac{\text{Distance}}{\text{Time}}$$

$$r = \frac{d}{t}$$

$$= \frac{100 \text{ mi} \cdot 2}{5 \text{ hrs}} = 40 \text{ mph}$$

USING THE RATE FORMULA

Example Theresa made the trip to her mother's house in $2\frac{1}{2}$ hours. If the distance she drove is 100 miles, what was her average speed (rate)?

Step 1

Rewrite the distance formula to find the rate.

$$\frac{d}{t} = r$$

Step 2

Substitute the known values for distance and time.

$$r = 100 \div 2\frac{1}{2}$$

Step 3

Divide to find the rate.

$$\frac{20}{1} \times \frac{2}{5} = 40 \text{ miles per hour}$$

Theresa drove at an average speed of **40 miles per hour**.

Find the average speed (rate) in the problems below. Round to the nearest mile per hour.

2. distance = 120 miles

time = 3 hours

rate = _____ mph

$$r = d/t$$

$$r = \frac{120 \text{ mi}}{3 \text{ hrs}}$$

$$r = 40 \text{ mph}$$

distance = 152 miles

time = $3\frac{1}{2}$ hours

rate = _____ mph

$$r = d/t$$

$$r = \frac{152 \text{ mi}}{3\frac{1}{2} \text{ hrs}}$$

$$= \frac{152 \text{ mi}}{7/2 \text{ hrs}}$$

$$= 152 \text{ mi} \left(\frac{2}{7} \text{ hrs} \right)$$

$$= \frac{304 \text{ mph}}{7}$$

$$= 43\frac{3}{7}$$

$$r = 43 \text{ mph}$$

Estimating Trick

$$150 \times 2 = 300$$

distance = 276 miles

time = $4\frac{3}{4}$ hours

rate = _____ mph

$$r = d/t$$

$$r = \frac{276 \text{ mi}}{4\frac{3}{4} \text{ hrs}}$$

$$= \frac{276 \text{ mi}}{19/4 \text{ hrs}}$$

$$= 276 \text{ mi} \left(\frac{4}{19} \text{ hrs} \right)$$

$$= 58.11$$

$$r = \cancel{58.11} \text{ mph}$$

TRICK

$$d \approx 275/5$$

$$\approx 55 \text{ mi}$$

if ≥ 5 , round up

if < 5 , round down

$$43\frac{3}{7} = 43.43$$

$$\text{and } .43 < .5$$

$\therefore$ round down

distance = 598 miles

time = 12 hours

rate = _____ mph

$$r = d/t$$

$$r = \frac{598 \text{ mi}}{12 \text{ hrs}}$$

$$= \frac{299 \text{ mi}}{6 \text{ hrs}}$$

$$= 49\frac{5}{6} \text{ mph}$$

$$r = 50 \text{ mph}$$

$$\div 7 > 40$$

Time

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$$\text{Time} = \frac{\text{Distance}}{\text{Rate}}$$

$$\underline{t = d/r} \quad \text{or}$$

$$r = d/t$$

$$t = \frac{120 \text{ mi}}{40 \text{ mph}}$$

$$t = 3 \text{ hrs}$$

$$120 \text{ mi} \div \frac{40 \text{ mi}}{1 \text{ hr}}$$

$$= 120 \cancel{\text{mi}} \cdot \frac{1 \text{ hr}}{40 \cancel{\text{mi}}}$$

$$t = 3 \text{ hrs}$$

USING THE TIME FORMULA

Example How long will it take Henri to travel 120 miles if he drives at an average speed of 40 miles per hour?

Step 1

Rewrite the distance formula to find the time.

$$\frac{d}{r} = t$$

Step 2

Substitute the known values for distance and rate.

$$t = 120 \div 40$$

Step 3

Divide to find the time.

$$\begin{array}{r} 3 \\ 40 \overline{) 120} \end{array} = 3 \text{ hours}$$

It will take Henri **3 hours** to travel 120 miles at 40 miles per hour.

Find the time in the problems below. Round to the nearest hour.

3 A. distance = 140 miles
rate = 35 mph
time = _____ hours

3 B. distance = 245 miles
rate = 40 mph
time = _____ hours

3 C. distance = 500 miles
rate = 55 mph
time = _____ hours

3 D. distance = 367 miles
rate = 32 mph
time = _____ hours

$$r = d/t$$

$$t = d/r$$

$$= 140 \text{ mi} \div \frac{35 \text{ mi}}{1 \text{ hr}}$$

$$= 140 \cancel{\text{mi}} \times \frac{1 \text{ hr}}{35 \cancel{\text{mi}}}$$

$$t = 4 \text{ hrs}$$

Because $35 \times 2 = 70$
 $70 \times 2 = 140$

Distance/Rate/Time Applications

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Solve these distance, rate, and time problems. Round your answer to the nearest whole number, if necessary.

4. Wendy drove for 6 hours to visit her friends. Her average speed was 48 miles per hour. How many miles did Wendy travel?
5. The Backmans moved to a town 1,575 miles from their old home. If it took the moving company 36 hours to get to their new home, what was the average speed of the moving van?
6. Ernesto drove 3 hours to a conference. His average speed was 53 miles per hour. How many miles did he drive?
7. Jan can drive to his brother's at an average speed of 32 miles per hour. The place is 30 miles away. How long does it take for Jan to get to his brother's house?
8. **Multiple Answers** Yosef drives 45 miles round-trip to and from work each day. How long does it take him (to the nearest *tenth* of an hour) to drive round-trip if he averages 25, 30, 35, 40, or 45 miles per hour depending on traffic?